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Twisted conductors on round core benchmark (3-D)

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Last modified:

September 1, 2026

The third simulation problem in this chapter consists of helically twisted HTS tapes as shown in Fig. 1. This 3-D benchmark, often referred as *Conductor on Round Tube* (CORT) or *Conductor on Round Core* (CORC®), represents a type of cable that offers many advantages for magnet designs. For example, they offer high mechanical strength and flexibility, as well as a low contact resistance between the tapes to facilitate current sharing around defects [1].

The modelled geometry is an idealized, single layer and infinitely long cable consisting of 3 HTS tapes helically wound around a non-conducting core. Fig. 1a shows the full simulated geometry and mesh, with the air domain and the HTS tapes. The outer air boundary considers the natural $\mathbf{n} \cdot \mathbf{B} = 0$ condition. Only one third of a full pitch of the cable is modelled, taking advantages of the periodic boundary conditions that connect the front and back faces. Fig. 1b highlights the periodicity of the tapes, matching the terminals via a translation along the z axis. Considering that each tape connects respectively $A_1 \rightarrow B_2$, $B_1 \rightarrow C_2$ and $C_1 \rightarrow A_2$, all the conducting domains are topologically within the same connected component, creating a single hole in the air domain (i.e. the same current flows through every tapes, which can be prescribed using a single cohomology cut).

The tapes are modelled using thin-shells, each containing 10 layers to highlight the current patterns in such cables, as discussed below. The specific geometry parameters of this benchmark are in Tab. 1. Every thin-shell layers are modelled using the same electrical property, which is a HTS resistivity that follows the power-law with a constant critical current density of $J_c = 4 \times 10^{10} \text{ A/m}^2$ and an exponent factor of $n = 25$. A 50 Hz AC current with an amplitude of 160 A is enforced in the tapes¹. In terms of the BELFEM features, this benchmark highlights the multilayered thin-shells and the use of periodic boundary conditions with the $\mathbf{H}$ - ϕ formulation.

Fig. 2 presents the current density distributions in the HTS tapes on the $z = 0$ cross

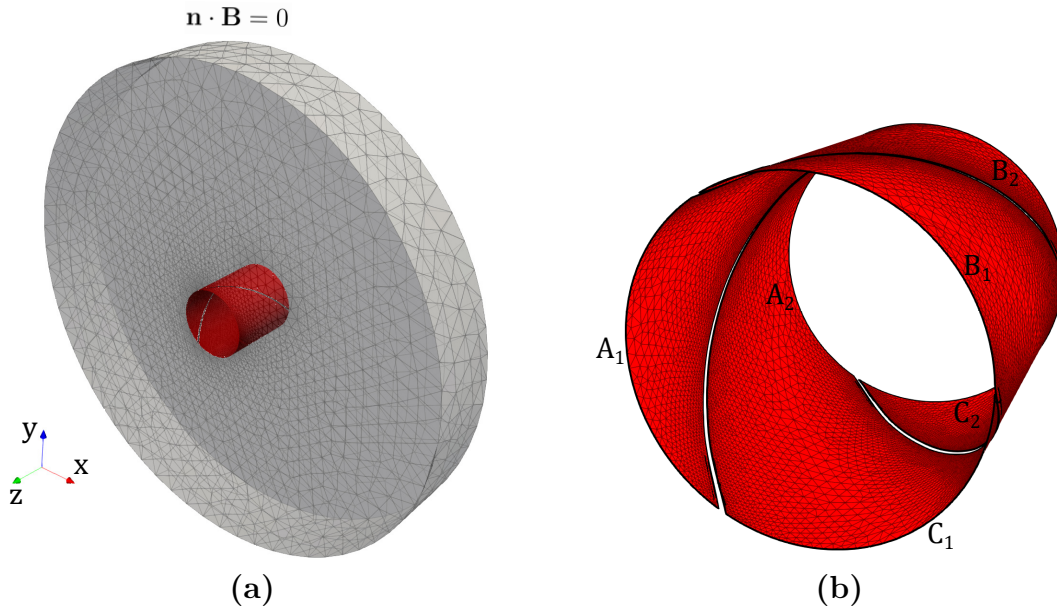


Figure 1: (a) Full geometry and mesh of the simulated twisted conductor problem with the anti-symmetry (perfect electric conductor) boundary condition at the outside boundary, and (b) zoom on the twisted tapes. The periodicity of the tapes is shown such that the matching letters (A_i , B_i and C_i) represent the matching tape periodic boundaries.

¹For a tape with a thickness of 1 μm , a width of 4 mm and a critical current density of $J_c = 4 \times 10^{10} \text{ A/m}^2$, the current amplitude of 160 A represents the short-sample critical current of the tapes.

Parameter	Value
Tapes width	4 mm
Tapes thickness	1 μm
Pitch	17 mm
Core radius	3 mm
Gap	100 μm
Pitch angle	45°

Table 1: Geometric and parameters of the 2-D undulator benchmark.

section at various times during the simulation. With this benchmark, we highlight the specific current pattern that arises in helical conductor configurations for undersaturated currents, known as the Garber effect [2, 3]. The main characteristic of this pattern is expressed by the direction of the current within the tapes. Towards the interior of the cable along the radius, the current is essentially azimuthal (i.e. $J \approx |J_\phi|$), while it is essentially axial (i.e. $J \approx |J_z|$) towards the exterior of the cable. This behaviour is caused by the fact that the magnetic field in the interior region of the cable is purely axial, while it is purely azimuthal in the exterior region. Combined with the minimal resistivities of the superconductors, the current path forms a specific zig-zag pattern within the tapes. This is highlighted in Fig. 2, where the first column presents J_z/J_c and the second column presents J_ϕ/J_c . The saturation of J_z occurs starting from the outer layers, while the saturation of J_ϕ from the inner layers, as the current increases with time. At $t = 3$ ms, both current fronts meet and the total current begins to orient in the direction of the helical winding of the tapes. This is when the current reaches

$$I_t = \frac{I_c}{\sin \alpha + \cos \alpha}, \quad (1)$$

where I_c is the short-sample critical current of the tapes and α is the pitch angle. This current is often referred as the current threshold in these cables, over which the losses drastically increases in the tapes.

We also note that, considering the pitch angle of 45°, the saturation of the tapes occurs symmetrically between both components in this specific example. However, an unsymmetrical current pattern between the axial and the azimuthal components are observed when modelling these twisted cables with other pitch angles.

It is worth noting how the thin-shell model, even though it is a numerical and geometrical approximation, is able capture such subtleties within the current distributions in HTS tapes, making it a physically robust model for a large number of applications.

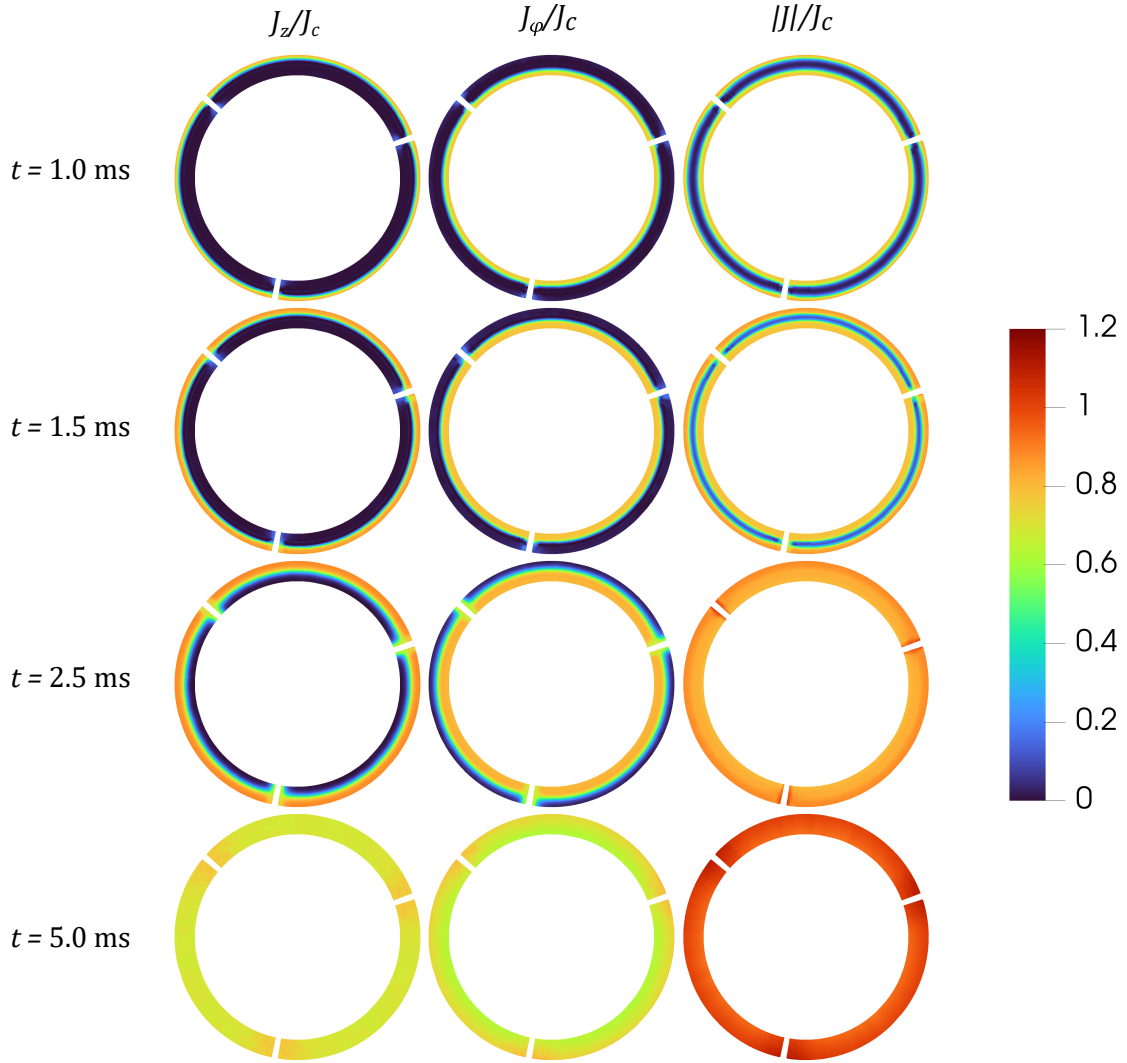


Figure 2: Normalized current density distributions on the $z = 0$ cross section of the twisted conductor benchmark containing 3 tapes with 10 thin-shell layers each. The z component, the ϕ component and the norm of the current density, normalized by J_c are presented at various time steps during the first current ramp to highlight the Garber effect before saturation. The thicknesses of the tapes are artificially increased to help the visualization.

1 References

- [1] D. C. Van der Laan, J. D. Weiss, and D. McRae, “Status of corc® cables and wires for use in high-field magnets and power systems a decade after their introduction,” *Superconductor Science and Technology*, vol. 32, no. 3, p. 033001, 2019.
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